Bernstein's "Center"

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1. The problem
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Let G be a reductive group over a non-archimedean local field and H the convolution algebra of locally constant functions with compact support on G. We determine the algebra of multipliers of H, in terms of the cuspidal representations of Levi subgroups of parabolic subgroups of G. The proof is based on a study of the category of algebraic representations (also called smooth) of G.

In §3, we obtain as corollaries various finiteness results.

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1. THE PROBLEM.

1.1. Let G be a totally disconnected locally compact group. Fix a field k of characteristic 0 and let $H_k(G)$, or simply $H(G)$, be the convolution algebra of locally constant measures with compact support — with coefficients in k — on G; for $k = \mathbb{C}$, "locally constant" means "locally a multiple of a Haar measure"; this is a purely algebraic notion, which allows one to work over an arbitrary k. If G has an open compact subgroup which is a pro-p-group, one can even take for k any $\mathbb{Z}[1/p]$-algebra. It matters little: the reader will lose little by assuming $k = \mathbb{C}$, and we will assume it anyway from 1.8 onwards.

Consider the following property of a ring H (which is not assumed to have a unit).

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(Id). For any finite family (x_i) of elements of H, there exists an idempotent e such that $ex_ie = x_i$ (i.e. such that $x_i \in eHe = eH \cap He$) for all i.

In [4], D. Flath calls "idempotented algebras" the $\mathbb{C}$-algebras satisfying (Id). Let I denote the set of idempotents of H. For $e, f \in I$, the following conditions are equivalent: $eHe \subset fHf$, $e \in fHf$ and $e = fef$. If they are satisfied, we write $e \ll f$. The relation $\ll$ is an order on I, and the axiom (Id) is equivalent to it being filtered, and that H is the (filtered) union of the eHe.

For any open compact subgroup K of G, the direct image under the inclusion of K in G of the normalized Haar measure of K is an idempotent e_K of $H(G)$. One thus obtains enough idempotents to verify (Id): $H(G)$ satisfies (Id), and the set of the e_K is cofinal in the set I of idempotents of $H(G)$.

An algebraic representation (V, π) of G is a k-vector space V, equipped with a (k-linear) action of G such that the stabilizer of every vector $v \in V$ is open (terminology of [1], [2]; F. Rodier [5] says "smooth"). This notion is equivalent to that of a non-degenerate $H(G)$-module V: such that $H(G) \cdot V = V$, i.e. such that for every $v \in V$ there exists an idempotent e such that $ev = v$. The dictionary is: for $v \in V$ to be fixed by K, an open compact subgroup of G, it is necessary and sufficient that $e_K v = v$. For λ in k and g in G, we then have $(\lambda \cdot \delta_g * e_K)v = \lambda\pi(g)v$.

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We will denote by (Alg G) the category of algebraic representations of G. In the following, we will simply say representation for "algebraic representation".

1.2. Let H be a ring satisfying (Id). For $h \in H$, the endomorphism $v \mapsto hv$ of the underlying abelian group of a non-degenerate H-module V is functorial in V. If H does not have a unit, the forgetful functor ω : (non-degenerate modules) $\to$ (abelian groups) can have other endomorphisms. For example, $H(G)$ does not have a unit for G non-discrete, and for $g \in G$, $v \mapsto gv$ is a functorial endomorphism of the underlying k-vector space of the representation V.

For any idempotent e, we have $H = He \oplus H(1-e)$, where $H(1-e) := \{h - he \mid h \in H\}$ is the left annihilator of e. Let $\mathcal{T}$ be the topology of H having as a fundamental system of neighborhoods of 0 the $H(1-e)$. The completion $H^\wedge$ of H for $\mathcal{T}$ is the projective limit over I of the He, for the transition morphisms $Hf \to He : x \mapsto xe$: an element $\hat{h}$ of $H^\wedge = \varprojlim He$ is a system of elements $\hat{h}(e) \in He$, with $\hat{h}(e) = \hat{h}(f) \cdot e$ for $e \ll f$. Taking $g \in I$ which dominates e and f, one verifies that $\hat{h}(e) = \hat{h}(g) \cdot e$ as soon as $He \subset Hf$ (i.e., $e = ef$).

For any non-degenerate module V, equipped with the discrete topology, $H \times V \to V : (h, v) \mapsto hv$ is continuous, and $H^\wedge$ acts on V by continuity: if $ev = v$, $\hat{h} \cdot v = \hat{h}(e) \cdot v$.

Lemma 0.1 (1.2.1)**.** *We have $H^\wedge \xrightarrow{\sim}$ End(forgetful functor ω).*

Let H_s be the H-module H, for left multiplications. Since every non-degenerate H-module is a quotient of a multiple of H_s, it suffices to verify that every endomorphism of the abelian group H_s, which commutes with $\mathrm{End}_H(H_s)$ (and in particular with right multiplications) is defined by an element of $H^\wedge$. We have more precisely:

Lemma 0.2 (1.2.2)**.** *$H^\wedge$ embeds into $\mathrm{End}_{\mathbb{Z}}(H)$. Its image is the closure for the topology of simple convergence of the set of left multiplications. It is also the commutant of the right multiplications.*

The identity $\hat{h}(e) = \hat{h} \cdot e$ shows that $H^\wedge$ embeds into $\mathrm{End}_{\mathbb{Z}}(H)$, and that the topology induced by the simple topology is $\mathcal{T}$. Since composition is continuous for the simple topology, we have

$$H^\wedge \hookrightarrow \overline{\{\text{left multiplications}\}} \subset \text{commutant of right multiplications.}$$

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and the reader will verify that if φ is in the commutant, it is the image of $h^\wedge \in H^\wedge$ defined by $h(e) = \varphi(e)$.

1.3. Let us specify the multiplication in $H^\wedge$. Let h_1 and h_2 be in $H^\wedge$, and $e \in I$. There exists $f \in I$ such that $h_2(e) = fh_2(e)$, and $(h_1 \cdot h_2)(e)$ is

$$(h_1 \cdot h_2) \cdot e = h_1(h_2 e) = h_1 h_2(e) = h_1 f h_2(e) = h_1(f) \cdot h_2(e).$$

The multiplication is continuous.

1.4. Translation (of $H(G)^\wedge = \lim\operatorname{proj} H(G) \cdot e_k$).

$H(G)^\wedge$ is the space of distributions T on G such that for every open compact subgroup K of G, $T * e_k$ has compact support.

Our goal is, for G a reductive group over a local field, the computation of the center of $H(G)^\wedge$.

For H satisfying (Id), the center Z of $H^\wedge$ is the commutant of H in $H^\wedge$ (by continuity of multiplication) and thus 1.2.2 identifies with the commutant in $\operatorname{End}_{\mathbb{Z}}(H)$ of the left and right multiplications.

Lemma 0.3. *1.5. (i) The center Z of H is the ring of endomorphisms of the identity functor of the category of non-degenerate H-modules.*

(ii) It is the projective limit over I of the centers of eHe.

In order for $z \in H^\wedge = \operatorname{End}(\omega)$ to provide, for each non-degenerate module V, a module endomorphism of V, it is necessary and sufficient that it commutes with H. This proves (i).

If $eHe \subset fHf$ and z is central in fHf, we have $ez = ze = eze$, and if $x \in eHe$, $[eze, x] = [z, x] = 0$: eze is central in eHe. This gives meaning to (ii): the transition morphism is $z \mapsto ze = eze$, and it is a matter of showing that for $z \in H^\wedge$ to commute with H, it is necessary and sufficient that ze be in the center of eHe, for every $e \in I$. The necessity is verified as above. For sufficiency, we first verify that $ez = ze$: for every $f \geq e$, we have $(ez - ze)f = ezf - zfe = e(fzf) - (fzf)e$, and we use the fact that fzf is central in fHf. Once this is established, for z to commute with eHe, it is necessary and sufficient that $ze = eze$ be in the center of eHe, and we use the fact that H is the union of the eHe.

1.6. Translation (of 1.5. (ii)). The center of $H(G)^\wedge$ is the space of distributions T on G such that for every open compact subgroup K of G, $T * e_k$ has compact support, and is in the center of the algebra[1]

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de Hecke $H(G, K) := e_K * H(G) * e_K$.

1.7. Variant. The center of $H(G)^\wedge$ is the space of distributions T on G, invariant by conjugation, and such that for every compact open subgroup K of G, $T * e_K$ has compact support.

The invariance by conjugation of T is indeed equivalent to the fact that for every locally constant measure (resp. distribution) with compact support U, one has $T * U = U * T$.

1.8. If k' is an extension of k, one has $H_{k'}(G) = H_k(G) \otimes_k k'$. For K a compact open subgroup, the k'-algebra $e_K H_{k'}(G) e_K$, and its center, are similarly deduced from the k-algebra $e_K H_k(G) e_K$, and its center, by extension of scalars. This, and 1.5 (ii), control the dependence on k of the center of $H_k(G)^\wedge$. In what follows, we shall make the hypothesis

$$k = \mathbb{C} \text{ and } G \text{ is countable at infinity.} \tag{1.8.1}$$

The algebras $H(G, K)$ are then of countable dimension, and Schur's lemma is valid ([1] 2.11). In particular, if z is in the center Z of $H(G)^\wedge$, for every irreducible algebraic representation (V, π) of G, z acts on V by a scalar $z(\pi)$, and we thus associate to z a function on the set $G^\wedge$ of irreducible algebraic representations of G. The algebra morphism

$$Z \to (\text{functions on } G^\wedge) \tag{1.8.2}$$

is injective: every representation of G is a quotient of a multiple of $H(G)_s$. For every $h \neq 0$ in $H(G)$, there exists an irreducible (V, π) such that $\pi(h) \neq 0$ ([1] 2.12). The representation $H(G)_s$ thus embeds into a product of irreducible representations, and z is determined by the function $z(\pi)$. Our determination of Z will essentially be a description of its image by (1.8.2).

[1] [[ILLEGIBLE — REASON: The sentence ends abruptly at the bottom of the page.]]

1.9. The center $Z(A)$ of an abelian category A is the endomorphism ring of the identity functor of A: the data for every object A of an endomorphism z_A of A, such that for every morphism $f : A \to B$, one has $z_B f = f z_A$.

If the abelian category A is the product of abelian categories $(A_i)_{i\in I}$ one has $Z(A) = \prod Z(A_i)$. Suppose that the category A admits direct sums indexed by I, and that every morphism $f : X \to \oplus Y_i$ such

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that the $pr_i f$ are zero implies f is zero. This is the case for (Alg G). Then, giving a decomposition of A into a product as above is equivalent to giving full subcategories A_i of A ($i \in I$) such that

(a) For X in A_i and Y in A_j, we have $\mathrm{Hom}(X, Y) = 0$ if $i \neq j$.

(b) Every object X is a sum

$$X = \bigoplus_i X_i, \quad \text{with } X_i \text{ in } A_i. \tag{1.9.1}$$

The center of $H(G)^\wedge$ is that of the category (Alg G) (1.1) and 1.5(i). A first step in the determination of the center of $H(G)^\wedge$ will be the decomposition 2.10, implicit in [1], of (Alg G) into a product. The decomposition results to which the end of this section is devoted are preliminaries to 2.10.

1.10. Let W be a representation of G. Its algebraic dual, or simply dual, is the space $W^\sim$ of linear forms fixed by an open subgroup of G. If we equip W with the linear topology for which the $(1 - e_K)W$ (K an open compact subgroup) form a fundamental system of neighborhoods of 0, it is the topological dual:

$$W^\sim = \lim\mathrm{ind}(W/(1 - e_K)W)^\vee = \lim\mathrm{ind}(W^K)^\vee.$$

Recall that the following conditions are equivalent ([1] 2.40):

(a) The coefficients $\langle \zeta, gw \rangle$ ($w \in W, \zeta \in W^\sim$) are functions on G with compact support.

(b) For every open compact subgroup K and every $w \in W$, the function from G to W: $g \mapsto e_K g w$ is compactly supported.

If they are satisfied, and W is finitely generated, for example irreducible, W is <u>admissible</u> ([1] 2.41), i.e., the W^K are finite-dimensional. Condition (b) can also be stated:

(b*) For every $w \in W$, if $g \to \infty$ in G (according to the filter of complements of compact sets), we have $gw \to 0$ (for the linear topology above).

We denote by $(\mathrm{Alg}\ G)_f$ the category of representations of G satisfying (a) and (b).
1.11. Let (V,π) be irreducible in $(\mathrm{Alg}\ G)_f$, $(\mathrm{Alg}\ G)(\pi)$ be the category of representations that are sums of copies of (V,π), and $(\mathrm{Alg}\ G)(\mathrm{off}\ \pi)$ be that of representations without any subquotient isomorphic to (V,π). Parallel arguments...

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... to those of the theory of finite groups give

$$(\mathrm{Alg}\, G) = (\mathrm{Alg}\, G)(\pi) \times (\mathrm{Alg}\, G)(\mathrm{off}\ \pi) \tag{1.11.1}$$

([1] 2.44, at least for G unimodular) from which an idempotent $e(\pi)$ of the center of $H(G)^\wedge$, taking the value 1 on $(\mathrm{Alg}\, G)(\pi)$ and 0 on $(\mathrm{Alg}\, G)(\mathrm{off}\ \pi)$.

Suppose G is unimodular. Only this case will matter to us. The proof of (1.11.1) gives for the distribution $e(\pi)$ (1.7) the following expression, which will not be used by us. Let d_π be the formal degree of π (intrinsically, a Haar measure on G) and θ_π be the character of π (a generalized function). We have $\overline{\theta}_\pi(g) = \theta_\pi(g^{-1})$, because π is unitarizable, and

$$e(\pi) = \theta_\pi(g^{-1}) \cdot d_\pi\ . \tag{1.11.2}$$

<u>Variants.</u> Let A be a set of isomorphism classes of irreducible representations in $(\mathrm{Alg}\, G)_f$, and $(\mathrm{Alg}\, G)(\mathrm{off}\ A)$ the category of representations without subquotients of isomorphism classes in A.

a) If A is finite, we have

$$(\mathrm{Alg}\, G) = \left(\bigoplus_{\pi\in A}(\mathrm{Alg}\, G)(\pi)\right) \times (\mathrm{Alg}\, G)(\mathrm{off}\ A)\ . \tag{1.11.3}$$

b) The same decomposition holds for an infinite family, if for each open compact subgroup K there are in A only a finite number of (V,π) such that $V^K \neq 0$. It is necessary to verify 1.9 (b). Let then W be a representation of G. Let $W(\pi)$ be the component (1.11.1) of W in $(\mathrm{Alg}\, G)(\pi)$ $(\pi \in A)$. For K an open compact subgroup of G and $A' \subset A$ a finite subset of A containing all (V,π) in A such that $V^K \neq 0$, the decomposition (1.9.1) deduced from (1.11.3) for A': $W = \bigoplus_{\pi\in A'} W(\pi) \times W(\mathrm{off}\ A')$ induces a decomposition

$$W^K = \bigoplus_{\pi\in A'} W(\pi)^K \times W(\mathrm{off}\ A')^K = \bigoplus_{\pi\in A} W(\pi)^K \times W(\mathrm{off}\ A')^K\ ,$$

with $W(\mathrm{off}\ A')^K$ independent of A'. Let us set $W(\mathrm{off}\ A)^K := W(\mathrm{off}\ A')^K$ (A' sufficiently large rel. K) and $W(\mathrm{off}\ A) = \bigcup_K W(\mathrm{off}\ A)^K$. By passing to the limit over K being smaller and smaller, we further have

$$W = \bigoplus_{\pi\in A} W(\pi) \times W(\mathrm{off}\ A)$$

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and this decomposition, functorial in W, gives (1.11.3).

c) Without a finiteness hypothesis on A, an argument parallel to b) shows that every admissible representation admits a unique decomposition

$$W = \bigoplus_{\pi\in A} W(\pi) \times W(\mathrm{outside}\ A).$$

This decomposition still holds for every filtered inductive limit of admissible representations, and in particular for every representation in $(\mathrm{Alg}\ G)_f$ (recall that every W in $(\mathrm{Alg}\ G)_f$ of finite generation is admissible). Taking for A the set $\widehat{G}_f$ of all isomorphism classes of irreducibles in $(\mathrm{Alg}\ G)_f$, we find that

$$(\mathrm{Alg}\ G)_f = \prod_{\pi\in\widehat{G}_f} (\mathrm{Alg}\ G)(\pi). \tag{1.11.4}$$

If $\widehat{G}_f$ satisfies the finiteness condition b), we have furthermore

$$(\mathrm{Alg}\ G) = (\mathrm{Alg}\ G)_f \times (\mathrm{Alg}\ G)(\text{outside } \widehat{G}_f). \tag{1.11.5}$$

1.12. Let M be a free $\mathbb{Z}$-module of finite type and $u : G \to M$. We suppose that the restriction of u to the center $Z(G)$ of G: $Z(G) \to M$ has a compact kernel and a finite cokernel. Let $G^\circ := \mathrm{Ker}(u)$ and (V°, π°) be irreducible in $(\mathrm{Alg}\ G^\circ)_f$. Let G_1 be the subgroup of G formed of g such that π° is isomorphic to its conjugate by g. It contains G° and the center of G, hence is of finite index, and the set A of isomorphism classes of conjugates of (V°, π°) is finite. For every representation (V, π) of G, (1.11.3) applied to the restriction of V to G° provides a decomposition $V = V' \oplus V''$, where V' is the sum of the sub-representations isomorphic to a conjugate of (V°, π°). This decomposition is stable under G, and provides a decomposition

$$(\mathrm{Alg}\ G) = (\mathrm{Alg}\ G)_A \times (\mathrm{Alg}\ G)_{\text{outside } A}. \tag{1.12.1}$$

We propose to describe the category $(\mathrm{Alg}\ G)_A$.

(1.12.2) For W in $(\mathrm{Alg}\ G)_A$, let W_1 be the subspace of W that is the sum of the sub-G°-representations of G° isomorphic to (V°, π°). It is stable under G_1. The functor $W \mapsto W_1$ is an equivalence from $(\mathrm{Alg}\ G)_A$ to the category $(\mathrm{Alg}\ G_1)_{\pi^\circ}$ of representations of G_1 whose restriction to G° is a sum of representations isomorphic to π°.[2]

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multiple of (V°, π°). The inverse equivalence is the induction from G_1 to G°.

(1.12.3) Let $\tilde{G}_1$ be the group of pairs (g, P) with $g \in G_1$ and $P : V^\circ \xrightarrow{\sim} V^\circ$ such that $P\pi^\circ(h)P^{-1} = \pi^\circ(ghg^{-1})$. It is a central extension of G_1 by $\mathbb{C}^*$, and G° is identified with a subgroup of $\tilde{G}_1$:

$$\begin{tikzcd}
 & G^\circ \arrow[d, "g", "(g,\pi^\circ(g))"'] & \\
\mathbb{C}^* \arrow[r, "z\mapsto(e,z)"'] & \tilde{G}_1 \arrow[r, "(g,P)\mapsto g"'] & G_1
\end{tikzcd}$$

The representation π° of G° extends to $\tilde{G}_1$ by $(g, P) \mapsto P$.

The quotient $\tilde{M}_1$ of $\tilde{G}_1$ by the normal subgroup G° is a central extension of the image M_1 of G_1 in M by $\mathbb{C}^*$. The center $\tilde{C}$ of $\tilde{M}_1$ is the inverse image of a subgroup C of M_1 containing the image of the center of G, and thus is of finite index.

The functor $H \mapsto H \otimes V_0$ is an equivalence from the category of vector spaces to that of representations of G° that are multiples of V_0. The inverse equivalence is $W \mapsto \mathrm{Hom}(V_0, W)$. Giving an action π of G_1 on $H \otimes V_0$, extending the given action of G°, is equivalent to giving an action ρ of $\tilde{M}_1$ on H, such that $z \in \mathbb{C}^* \subset H$ acts by multiplication by z^{-1}: letting the image in $\tilde{M}_1$ of $(g, P) \in \tilde{G}_1$ act by Q such that $\pi(g) = Q \otimes P$:

$$\pi(g) = \rho(g, P) \otimes P$$

This construction is an equivalence from $(\mathrm{Alg}\ G^1)_{\pi^\circ}$ to the category $\mathcal{B}$ of representations of $\tilde{M}_1$, inducing $z \mapsto z^{-1}$ on $\mathbb{C}^*$. The equivalences $(\mathrm{Alg}\ G)_A \sim (\mathrm{Alg}\ G_1)_{\pi^\circ} \sim \mathcal{B}$ induce an isomorphism from the center $Z((\mathrm{Alg}\ G)_A)$ to $Z(\mathcal{B})$. The category $\mathcal{B}$ is well known (Heisenberg group theory) and, after some reminders on its structure, we will deduce a description of $Z((\mathrm{Alg}\ G)_A)$ from it.

1.13. The commutator map induces on $\tilde{M}_1/\tilde{C} = M/C$ a non-degenerate alternating bimultiplicative map $M/C \times M/C \to U_1 \subset \mathbb{C}^*$. It is known that every finite group H, equipped with such a map $\langle \cdot, \cdot \rangle$, can be written

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$H = X \times X'$, where X' is the Pontryagin dual of X and where $\langle (x, x'), (y, y') \rangle = \langle x, y' \rangle \langle y, x' \rangle^{-1}$.. In particular, its order is a square.

[2] [[UNCERTAIN: The last part of the sentence "de restriction à G°" is followed by a notation that seems to indicate the type of restriction. — CONFIDENCE: 90% — REASON: Technical context of Clifford theory.]]

Let X be a subgroup of M/C, maximal among those on which the commutator is trivial. For example: corresponding to a decomposition as above. Its inverse image $\tilde{X}$ in $\tilde{M}_1$ is a maximal commutative subgroup.

It is known that for each character χ of $\tilde{C}$, such that $\chi(z) = z^{-1}$ for $z \in \mathbb{C}^*$, there exists, up to isomorphism, a unique irreducible representation of $\tilde{M}_1$ with central character χ. One obtains it by extending—in any way—χ to $\tilde{X}$ and by inducing from $\tilde{X}$ to $\tilde{M}_1$. If π is an irreducible representation, with $\pi(z) = z^{-1}$ for $z \in \mathbb{C}^*$, the others are therefore of the form $\pi\omega$, where ω is a character of M_1, and the isomorphism class of $\pi\omega$ depends only on $\omega|_C$.

Let us denote by $\mathbb{C}[\tilde{M}_1; z^{-1}]$ the $\mathbb{C}$-algebra generated by elements δ_m ($m \in \tilde{M}_1$), with the relations $\delta_m \cdot \delta_n = \delta_{mn}$ and $\delta_z = z^{-1} \cdot \delta_e$. If the $\tilde{m}$ form a set-theoretic lift of M_1 into $\tilde{M}_1$; it admits the $\delta_{\tilde{m}}$ as a vector space basis. Giving a $\mathbb{C}[\tilde{M}_1; z^{-1}]$-module is equivalent to giving a representation of $\tilde{M}_1$ where $z \in \mathbb{C}^*$ acts by z^{-1}: make δ_m act as m. The algebra $\mathbb{C}[\tilde{M}_1; z^{-1}]$ having a unit, the center of the category of its modules is simply its center, namely $\mathbb{C}[\tilde{C}; z^{-1}]$: it is simply generated by the center of the group $\tilde{M}_1$.

Let us translate into terms of representations.

1.14. Let T be the torus (in the sense of algebraic groups) with character group M. Its points are

$$T(\mathbb{C}) = \mathrm{Hom}(M, \mathbb{C}^*)$$

and for the algebra of regular functions, the group algebra $\mathbb{C}[M]$ of the group M. For $M = \mathbb{Z}^n$, this is the algebra of Laurent polynomials $\mathbb{C}[t_1, t_1^{-1}, \ldots, t_n, t_n^{-1}]$. Let $F \subset T(\mathbb{C})$ be the orthogonal of $C \subset M_1 \subset M$. The quotient torus T/F is the torus with character group C.

The set $\mathrm{Irr}(B)$ of isomorphism classes of irreducible representations of $\tilde{M}_1$, with $z \mapsto z^{-1}$, is, as we saw in 1.13, a principal homogeneous space under $(T/F)(\mathbb{C}) = \mathrm{Hom}(M, \mathbb{C}^*)/F = \mathrm{Hom}(C, \mathbb{C}^*)$: if (V, π) is irreducible in B, every irreducible is isomorphic to $(V, \pi\chi)$, for $\chi \in \mathrm{Hom}(M, \mathbb{C}^*)$, and $\pi\chi$ is isomorphic to $\pi\chi'$ if and only if

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$\chi|C = \chi'|C$. This endows $\mathrm{Irr}(\mathcal{B})$ with an algebraic variety structure, a function f on $\mathrm{Irr}(\mathcal{B})$ being regular if and only if the function $\chi \mapsto f(\pi_\chi)$ on $T(\mathbb{C})$ is regular. By passing to the quotient, it then comes from a regular function on T/F, i.e., from a finite linear combination of functions $\chi \mapsto \chi(c)$ ($c \in C$).

An element z of the center of $\mathcal{B}$ provides a function $z(\pi)$ on $\mathrm{Irr}(\mathcal{B})$: to (V, π), attach the scalar by which z acts on V. The description 1.13 of the center of $\mathcal{B}$: $Z(\mathcal{B}) = \mathbb{C}[\tilde{C}]^{F}$[3] can be reformulated as follows: the map $z \mapsto$ (function $z(\pi)$) is an isomorphism of $Z(\mathcal{B})$ with the ring of regular functions on the algebraic variety $\mathrm{Irr}(\mathcal{B})$.

The category equivalences of 1.12

$$(\mathrm{Alg}\ G)_A \sim (\mathrm{Alg}\ G^1) \underset{\pi}{\sim} \mathcal{B}$$

are compatible with torsion by a character of M. By translation one finds that $\mathrm{Irr}(\mathrm{Alg}\ G)_A$ is a homogeneous space under $T(\mathbb{C})$, and a principal homogeneous space under $(T/F)(\mathbb{C})$, hence an algebraic variety structure on $\mathrm{Irr}(\mathrm{Alg}\ G)_A$. Each element z of the center of $(\mathrm{Alg}\ G)_A$ defines a function $z(\pi)$ on $\mathrm{Irr}(\mathrm{Alg}\ G)_A$: $\pi \mapsto$ the scalar by which z acts, and one has

Proposition 0.4 (1.15)**.** *The map $z \mapsto$ (function $z(\pi)$) is an isomorphism of the center of the abelian category $(Alg\ G)_A$ with the ring of regular functions on the algebraic variety $Irr(Alg\ G)_A$.*

1.16. The fact that for π irreducible in $(\mathrm{Alg}\ G)_A$ and z in the center of $(\mathrm{Alg}\ G)_A$, the function $\chi \mapsto z(\pi_\chi)$ on $T(\mathbb{C}) = \mathrm{Hom}(M, \mathbb{C}^*)$ is regular can be deduced from the fact that π_χ depends "algebraically" on χ. Let us explain what this means, and how to use it.

Soit S be an affine algebraic variety, with ring of regular functions B. One defines an algebraic family of admissible representations of G, parametrized by S, as being a B-module V, endowed with an action of G (commuting with that of B), flat and for which the following property is satisfied:

(B-adm) For every open compact subgroup K of G, V^K is a B-module of finite type.

[3] [[UNCERTAIN: $\mathbb{C}[\tilde{C}]^F$ — CONFIDENCE: 70% — REASON: The character inside the brackets and the superscript are somewhat blurry; it could also be interpreted as $\mathbb{C}[C, z^{-1}]$.]]

The conjunction of the conditions "flat" and (B-adm) is equivalent to: each

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V^K is a projective B-module of finite type. If condition (B-adm) is satisfied, for every point $s \in S$, corresponding to a homomorphism $\sigma : b \mapsto b(s) : B \to \mathbb{C}$, the representation $V_s := V \otimes_{B,\sigma} \mathbb{C}$ of G is admissible.

Let us show that, for (V, π) irreducible in $(\text{Alg } G)_A$, the (V, π_χ) form an algebraic family, in the previous sense. We take $B = \mathbb{C}[M]$, the ring of regular functions on T. We have a "universal" character $\chi_{un} : M \to B$, such that for $\chi \in T(\mathbb{C})$ corresponding to $x : B \to \mathbb{C}^*$, we have $x\chi_{un} = \chi$. Let G act on $V_B := V \otimes B$ by $\pi \cdot \chi_{un}$. This is the algebraic family sought: we have $(V_B, \pi \cdot \chi_{un}) \otimes_{B,x} \mathbb{C} = (V, \pi_\chi)$.

The lemma below shows that z acts in $(V_B, \pi \cdot \chi_{un})$ by multiplication by an element $b \in B$. By functoriality, z acts in (V, π_χ) by $b(\chi) : z(\pi_\chi) = b(\chi)$ is a regular function of $\chi \in T(\mathbb{C})$.

Lemma 0.5 (1.17)**.** *Let S, B be as above, and V an algebraic family of admissible representations of G, parameterized by S. Suppose that there exists a set Ξ of points of S : $\Xi \subset S(\mathbb{C})$ such that the maps $b \mapsto b(s)$ $(s \in \Xi)$ separate the $b \in B$ ($\iff$ S is reduced and Ξ is Zariski-dense) and that the V_s are irreducible. Then every B-linear endomorphism z of V that commutes with the action of G is multiplication by an element $b \in B$.*

Proof. Let K be an open compact subgroup. By hypothesis, V^K is a projective B-module of finite type. Let $z(K)$ be the endomorphism of V^K induced by z. Schur's Lemma for the V_s $(s \in \Xi)$ implies that for every $\sigma : B \to \mathbb{C}$ corresponding to $s \in \Xi$, $z(K)$ acts as a scalar on $V^K \otimes_{B,\sigma} \mathbb{C}$. It follows that $z(K)$ is multiplication by an element $b \in B$: by localization, one reduces to assuming that the projective module V^K is free. The endomorphism $z(K)$ is then represented as a square matrix (z_{ij}). The $\sigma((z_{ij}))$ are scalar matrices. Since the σ $(s \in \Xi)$ separate the $b \in B$, (z_{ij}) is scalar: we have $z_{ii} = z_{jj}$ and $z_{ij} = 0$ for $i \neq j$. This, being true for every K, proves the lemma. □

We will need to reuse the algebraic family (V, π_χ) of representations of G introduced in 1.16. Formally: the representation $(V \otimes \mathbb{C}[M], \pi \otimes \chi_{un})$. It parameterizes all the irreducibles in $(\text{Alg } G)_A$ (with repetitions if $F \neq \{1\}$). Moreover

Lemma 0.6 (1.18)**.** *Every representation in $(Alg\ G)_A$ is a quotient of a multiple*

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of $(V \otimes \mathbb{C}[M], \pi \otimes \chi_{\text{un}})$.

The representation $V \otimes \mathbb{C}[M]$ is the induced representation from G° to G of the restriction of V to G°, in the sense that the restriction to $V \hookrightarrow V \otimes \mathbb{C}[M] : v \mapsto v \otimes \delta_e = v \otimes 1$ provides an isomorphism

$$\text{Hom}_G(V \otimes \mathbb{C}[M], X) = \text{Hom}_{G^\circ}(V, X).$$

The lemma thus follows from the fact that X in $(\text{Alg } G)_A$ has a restriction to G° which is a quotient of a sum of copies of $V|_{G^\circ}$.

Other proof. One verifies that $\mathbb{C}[\tilde{M}_1; z^{-1}]$ is an Azumaya algebra over its center $\mathbb{C}[\tilde{C}; z^{-1}]$. To prove 1.18, it then suffices to show that the $\mathbb{C}[\tilde{M}_1; z^{-1}]$-module corresponding by 1.12 to $(V \otimes \mathbb{C}[M], \pi \otimes \chi_{\text{un}})$ is free $\neq 0$ as a $\mathbb{C}[\tilde{C}; z^{-1}]$-module.

Remark. If $\tilde{M}_1$ is not commutative, there is no algebraic parametrization without repetition of the irreducibles of $(\text{Alg } G)_A$. This follows from the fact that, if L is the fraction field of the center $\mathbb{C}[\tilde{C}; z^{-1}]$ of $\mathbb{C}[\tilde{M}_1; z^{-1}]$, $\mathbb{C}[\tilde{M}_1; z^{-1}] \otimes_{\mathbb{C}[\tilde{C}; z^{-1}]} L$ is a division algebra over L.

1.19. Suppose G is unimodular, and $u : G \to M$ is surjective. Let $n \cdot \sum \pi_i$ be the decomposition of $\pi|_G$ into irreducible representations, d_i the formal degree of π_i (Haar measures on G°, which are moreover equal to each other) and dg the Haar measure on G with restriction to G° being $n \cdot \sum d_i$. Let $m \in M$ be orthogonal to F. We have (1.5 (i), 1.9, (1.12.1))

$$Z(H(G)^\wedge) \sim Z(\text{alg } G) \sim Z((\text{Alg } G)_A) \times Z((\text{Alg } G)_{\text{outside } A});$$

let $z(\pi, m)$ be the element with image $\pi_\chi \to \chi(m)$ in $Z((\text{Alg } G)_A)$ (1.18) and 0 in $Z((\text{Alg } G)_{\text{outside } A})$.

One verifies that, viewed as a distribution on G (1.7), $z(\pi, m)$ is given by the formula

$$\begin{aligned} z(\pi, m) &= \frac{1}{\#F}\theta_\pi(g^{-1})\, dg|_{u^{-1}(m)} \\ &= \frac{1}{\#F}\left(\int \theta_{\pi_\chi}(g^{-1})\chi(m)\, d\chi\right) dg, \end{aligned}$$

where the integration is over the compact group of unitary characters of M, equipped with its normalized Haar measure.

1.20. A representation W of G is said to be quasi-cuspidal if the following equivalent conditions are satisfied.

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(a) The coefficients of the restriction of W to G° have compact support.

(b) The coefficients of W have compact support mod the center $Z(G)$ of G.

(c) For all $w \in W$, if $g \to \infty$ in G, according to the filter of complements of compact sets mod $Z(G)$, then $gw \to 0$ in W, for the topology of $(1 - e_K)W$ (cf. 1.10 (b*)).

We have (a) $\Leftrightarrow$ (c) for G° (1.10) and ((c) for G°) $\Rightarrow$ (b) $\Rightarrow$ (a). We say that W is cuspidal if it is quasi-cuspidal and admissible.

Let $(\mathrm{Alg}\ G)_{\mathrm{qc}}$ be the category of quasi-cuspidal representations of G. For each orbit Ω of G in the set $(G^\circ)^\wedge_{\mathrm{f}}$ of isomorphism classes of irreducibles in $(\mathrm{Alg}\ G^\circ)_{\mathrm{f}}$, let us denote provisionally $(\mathrm{Alg}\ G)_\Omega$ the category of representations of G whose restriction to G° is a sum of representations in Ω. It follows from (1.11.4) that

$$(\mathrm{Alg}\ G)_{\mathrm{qc}} = \prod_\Omega (\mathrm{Alg}\ G)_\Omega.$$

According to 1.13, the irreducibles in $(\mathrm{Alg}\ G)_{\mathrm{qc}}$, being irreducible in one of the $(\mathrm{Alg}\ G)_\Omega$, are admissible (i.e., quasi-cuspidal + irreducible $\Rightarrow$ cuspidal), and the orbits D of $T(\mathbb{C}) = \mathrm{Hom}(M, \mathbb{C}^*)$ acting by torsion $\pi \to \pi\chi$ on the set $\mathrm{Irr}((\mathrm{Alg}\ G)_{\mathrm{qc}})$ of isomorphism classes of irreducibles in $(\mathrm{Alg}\ G)_{\mathrm{qc}}$ thus correspond bijectively to the orbits Ω: to the orbit $\{\pi\chi \mid \chi \in T(\mathbb{C})\}$ attach Ω such that π is in $(\mathrm{Alg}\ G)_\Omega$; Ω is the set of isomorphism classes of the constituents of $\pi|G^\circ$. We will write $(\mathrm{Alg}\ G)(D)$ for $(\mathrm{Alg}\ G)_\Omega$. This is the category of representations of G all of whose irreducible sub-quotients are in D. We have

$$(\mathrm{Alg}\ G)_{\mathrm{qc}} = \prod_D (\mathrm{Alg}\ G)(D) \tag{1.20.1}$$

(product over the orbits of $T(\mathbb{C}) = \mathrm{Hom}(M, \mathbb{C}^*)$ in $\mathrm{Irr}((\mathrm{Alg}\ G)_{\mathrm{qc}})$).

If, for every open compact subgroup K of G°, G° has only a finite number of isomorphism classes of irreducible representations (V, π) in $(\mathrm{Alg}\ G^\circ)_{\mathrm{f}}$ such that $V^K \neq 0$, we deduce from (1.11.5) a decomposition

$$(\mathrm{Alg}\ G) = (\mathrm{Alg}\ G)_{\mathrm{qc}} \times (\mathrm{Alg}\ G)_{\mathrm{nqc}} \tag{1.20.2}$$

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Let us equip $\mathrm{Irr}((\mathrm{Alg}\, G)_{\mathrm{qc}})$ with the structure of an algebraic variety for which it is, as a scheme, the disjoint union of the orbits of $T(\mathbb{C})\mathrm{Hom}(M, \mathbb{C}^*)$, equipped with their structure 1.14. According to 1.9 and 1.18, the map $z \mapsto z(\pi)$ which to $(V, \pi) \in \mathrm{Irr}((\mathrm{Alg}\, G)_{\mathrm{qc}})$ attaches the scalar by which z acts on V, identifies $Z((\mathrm{Alg}\, G)_{\mathrm{qc}})$ with the algebra of regular functions on $\mathrm{Irr}((\mathrm{Alg}\, G)_{\mathrm{qc}})$ (i.e., by definition of a disjoint union, regular on each component). If the finiteness condition above is satisfied, we further have by (1.20.2)

$$Z(\mathrm{Alg}\, G) = Z((\mathrm{Alg}\, G)_{\mathrm{qc}}) \times Z((\mathrm{Alg}\, G)_{\mathrm{nqc}}) \quad . \tag{1.20.3}$$

2. THE THEOREM.

2.1. We shall make essential use of the results of Chapter II of [1], generalized to the case of an arbitrary reductive group. If we contented ourselves with considering admissible representations, other references could be given, but it is essential for us to use non-admissible algebraic representations — for example, and typically, the representation $(V_B, \pi_{X_{\text{un}}})$ of 1.16 (or the regular representation). In loc. cit., only the case of $GL(n, F)$ (F a non-archimedean local field) is explicitly considered. The modifications to be made to the proofs to treat the case of $\underline{G}(F)$, $\underline{G}$ reductive over F, are the following.

a) Introduce the parabolic subgroups, their Levi decomposition, and the roots explicitly, where their introduction is only implicit. Similarly for the Bruhat, Cartan, and Iwasawa decompositions.

b) Let A be a maximal split torus. It is necessary to have open compact subgroups K that are arbitrarily small and good (rel. A) in the following sense. Let P be a parabolic containing A, $L \cdot U$ its Levi decomposition, with $L \supset A$, and $L\bar{U}$ the opposite parabolic containing A. Let us denote by $[H]$ the image measure in G of the normalized Haar measure of a compact subgroup H. We want to have $K = (K \cap U)(K \cap L)(K \cap \bar{U})$. The inverse image of the Haar measure of G by the map $ul\bar{u} : U \times L \times \bar{U} \to G$ being $\delta_p(l)^{-1} du\, dl\, d\bar{u}$, and δ_p taking the value 1 on $K \cap L$, we then have $[K] = [K \cap \bar{U}] * [K \cap L] * [K \cap U]$. The proof of Jacquet (cf. [1] 3.16; see also 3.2) shows that for every admissible representation V of G, denoting by V_U the coinvariants of

U in V ($V_U = V/\langle\{uv - v \mid u \in U, v \in V\}\rangle$), we have

$$V^K \longrightarrow (V_U)^{K \cap L}.$$

Here is a method for constructing such subgroups. Let (x_i) be a system of local analytic coordinates centered at e on G, and Λ the Lie lattice of G on which the dx_i take values in the ring O_F of the valuation of F. If Λ is the sum of its intersections with $\mathrm{Lie}(\overline{U})$, $\mathrm{Lie}(L)$, and $\mathrm{Lie}(U)$, and if n is sufficiently large (depending on Λ and the coordinate system), the set of g with coordinates $\equiv 0 \pmod{P^n}$ is a subgroup K of the desired type. Additional properties of Λ ensure them for K for n large enough. Thus, if Λ is stable under a compact subgroup K_o, K will be normalized by K_o. Let $Z(A)^+$ be the set of a in the centralizer $Z(A)$ of A (compact mod A) such that $||\alpha(a)|| \leq 1$ for every root α of A in U. If $\Lambda \cap \mathrm{Lie}(U)$ is contracted by every $a \in Z(A)^+$, we will have $a(K \cap U)a^{-1} \subset K \cap U$ for $a \in Z(A)^+$. If $\Lambda \cap \mathrm{Lie}(L)$ is stable under $Z(A)^+$, $K \cap L$ will be so as well.

c) Let $P = LU$ be a minimal parabolic, with $L \supset A$, $A^+ = A \cap Z(A^+)$ and K as above, with $K \cap L$ stable under A, $a(K \cap U)a^{-1} \subset K \cap U$, and $a^{-1}(K \cap \overline{U})a \subset K \cap \overline{U}$ for $a \in A^+$. For $a, b \in A^+$, we have $([K]a[K])([K]b[K]) = [K]ab[K]$ and the $[K]a[K]$ thus generate a commutative subalgebra, of finite type, of $H(G, K)$. This subalgebra B is large, in the sense that $H(G, K)$ is the sum of a finite number of uBv, because there exists in G a compact subset Ω such that $G = \Omega A^+ \Omega$. This allows applying the reasoning of [1] chapter 4. When the Cartan decomposition of G has the form $G = K_o A^+ K_o$ (this is the case for G split), and K is invariant in K_o, we have for example $H(G, K) = \sum uBv$, the sum being over u, v of the form $[K]c[K]$, $c \in K_o/K$.

The arguments of [1], chapter 2 apply equally to central extensions G of G_o — the group of points of a reductive group over F — by a finite group. The role of the parabolics will be played by the inverse images of the parabolics of G_o. If P is the inverse image of P_o, the unipotent radical U_o of P_o has a unique lift U normalized by P; this lift will play the role of the unipotent radical of P. The inverse image of a maximal split torus has a commutative subgroup of finite index, and this suffices for the arguments c).

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The results that follow also hold for these central extensions. To simplify notation, we will however restrict ourselves to the case where G is the group of points of a reductive group $\underline{G}$ over a non-archimedean local field F.

2.3. For $\underline{G}$ reductive over F and $G = \underline{G}(F)$, we denote by G° the following open subgroup of G. Let $\underline{T}$ be the largest quotient of $\underline{G}$ that is a torus, and T° the maximal compact subgroup of $\underline{T}(F)$. We set $G^\circ :=$ inverse image of T°. The quotient G/G° is of finite index in $\underline{T}(F)/T^\circ$ and thus isomorphic to $\mathbb{Z}^r$, for r the split rank of $\underline{T}$. The center $Z(G)$ of G has as connected component of e a torus isogenous to $\underline{T}$, and therefore maps onto a subgroup of finite index of G/G°: we are in situation 1.12.

According to [1], chapter 4, for each open compact subgroup K, G° has only a finite number of isomorphism classes of irreducible representations V with coefficients of compact support such that $V^K \neq 0$. The results 1.20 are therefore applicable: each representation W of G admits a canonical decomposition

$$W = W_{qc} \oplus W_{nqc}, \tag{2.3.1}$$

where the coefficients of W_{qc} are of compact support modulo $Z(G)$ and W_{nqc} has no subquotient whose coefficients are of compact support modulo $Z(G)$.

2.4. Quasi-cuspidal representations, as defined in paragraph 1, admit the following characterization: they are the representations W such that for U the unipotent radical of a proper parabolic P, one has $W_U = 0$. See for example [1] 3.21, where the case of $GL(n)$ is treated, with a proof valid for any G.

In the decomposition $W = W_{qc} \oplus W_{nqc}$ of a representation of G, W_{qc} is therefore the largest subrepresentation X such that $X_U = 0$, for U as above. Since $X_U \hookrightarrow W_U$ (exactness of $X \mapsto X_U$), it is the kernel of

$$W \to \prod_{P \neq G} \mathrm{Ind}_P^G(W_U) \tag{2.4.1}$$

(extend the product to proper parabolics, taken up to conjugation; it actually suffices to take maximal proper parabolics).

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Furthermore, no non-trivial sub-quotient of an induced representation is quasi-cuspidal: given the decomposition $(\mathrm{Alg}\ G) = (\mathrm{Alg}\ G)_{qc} \times (\mathrm{Alg}\ G)_{nqc}$, it would have a non-trivial direct factor in $(\mathrm{Alg}\ G)_{qc}$, and this would contradict the fact that for every sub-representation X of an induced [representation] $\mathrm{Ind}_P^G(X')$, X injects into $\mathrm{Ind}_P^G(X_U)$[4]. Induced representations are therefore in $(\mathrm{Alg}\ G)_{nqc}$.

2.5. Let P be a parabolic subgroup of G, U its unipotent radical, $L = P/U$ and δ_P the modulus function $p \mapsto \|\det(\mathrm{ad}\ p, \mathrm{Lie}\ U)\|$. The functor i_P^G is the induction functor, modified by $\delta_P^{1/2}$, from representations of L to those of G: to a representation (V, ρ) of L one associates one [representation] of P (of which L is a quotient), then a G-equivariant sheaf V on G/P, with fiber at e identified with V, one twists it by the sheaf of locally constant densities of weight $1/2$ and one passes to global sections. Once Haar measures on G and P have been chosen, it is equivalent to define $i_P^G(V)$ as the representation of G by right translations on the locally constant functions $f : G \to V$ satisfying for $p \in P$

$$f(pg) = \delta_P(p)^{1/2}\rho(p)f(g).$$

Let r_P^G be the functor of coinvariants under U, twisted by $\delta_P^{-1/2}$:

$$r_P^G(V) = V_U \otimes \delta_P^{-1/2}.$$

It is the left adjoint of i_P^G. One can reformulate (2.4.1) as

$$W_{qc} = \mathrm{Ker}(W \longrightarrow \prod_{P \neq G} i_P^G r_P^G W) \tag{2.5.1}$$

(product over proper parabolics, up to conjugacy. It suffices to take maximal proper parabolics).

The functors i_P^G and r_P^G are exact, and have the following transitivity property: for $Q(L)$ a parabolic of L, with inverse image Q in P, we have $i_P^G i_{Q(L)}^L = i_Q^G$ and $r_{Q(L)}^L r_P^G = r_Q^G$.

Let B be a $\mathbb{C}$-algebra, and consider representations equipped with a B-module structure (commuting with the action of G). We deduce from the exactness of i_P^G and r_P^G that these functors commute with $\otimes_B M$, for M a B-module. They therefore transform flat B-modules into flat B-modules.

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Torsion by $\delta_P^{1/2}$ introduced in these definitions has the advantage that i_P^G transforms unitary representations into unitary representations. For P and Q two parabolics, it ensures that the functor $r_Q^G i_P^G$ from representations of P/R_uP into Q/R_uQ is an iterated extension of the functors $i \circ r$ described below, without corrective torsion (see [2] 2.11). The required functors are indexed by $Q\backslash G/P$, i.e., by the relative positions of P and Q. For the trivial double class, we take the following functor. Let A be a maximal split torus in $P \cap Q$, and L, M the Levi subgroups of P, Q containing A. The intersection $L \cap M$ is a Levi subgroup of both the parabolic $Q \cap L$ of L and the parabolic $P \cap M$ of M, and we take $i_{P\cap M}^M r_{Q\cap L}^L$. For the double class of w, the corresponding functor is composed of $\mathrm{int}(w)$ and the previous functor for wPw^{-1} and Q.

The torsion by $\delta^{1/2}$ has however the disadvantage of introducing a parasitic irrationality: $\sqrt{q}$, where q is the number of elements of the residue field, which would become troublesome if one wanted to work with $\mathbb{Q}$-representations, rather than with $\mathbb{C}$-representations.

2.6. Let L be a Levi subgroup ($L = G$ is allowed), D a connected component of the set of isomorphism classes of irreducible cuspidal representations of L and Ω the corresponding orbit of L in the set of isomorphism classes of irreducible representations of L^0 with coefficients with compact support (1.20). Recall that $(\mathrm{Alg}\ L)(D)$ is the subcategory of $(\mathrm{Alg}\ L)$ formed by representations whose restriction to L^0 is a sum of copies of representations in Ω (1.20).

[4] [[UNCERTAIN: X_U — CONFIDENCE: 80% — REASON: The subscript U is slightly blurred but appears consistent with the font used for the unipotent radical U.]]

Lemma 0.7. *2.7. If a representation W of G is such that for every (L, D) as above and every parabolic P with Levi L, the component in (Alg L)(D) of $r_P^G W$ is zero, then $W = 0$.*

If $W \neq 0$, there exists a minimal P ($P = G$ is allowed) such that $r_P^G W \neq 0$. Let L be the Levi of P. The representation $r_P^G W$ has all its r_Q^L (Q a proper parabolic of L) zero, hence is in (Alg $L)_{qc}$ (2.4), and there exists D such that $r_P^G W$ has a non-zero component in (Alg $L)(D)$ (1.20.1).

Proposition-definition 2.8. (Alg $G)(L, D)$ is the subcategory of (Alg G) formed by representations W satisfying the following equivalent conditions:

(i) W embeds into a sum, over P with Levi L, of representations $i_P^G(W_P)$, with W_P in (Alg $L)(D)$.

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(ii) W is a sub-quotient of a sum of representations as in (i).

(iii) For any (M, E) as in 2.6, not conjugate to (L, D), and the parabolic Q with Levi M, the component in (Alg $M)(E)$ of $r_Q^G W$ is zero.

(iv) For P a parabolic with Levi L, $r_P^G W$ is in the sum of the (Alg $L)(E)$, for (L, E) conjugate to (L, D). Let $(r_P^G W)(D)$ be the component of $r_P^G W$ in (Alg $L)(D)$.

The natural map $\phi : W \longrightarrow \bigoplus_P i_P^G(r_P^G W)(D)$ is injective.

It is clear that (iv) $\Rightarrow$ (i) $\Rightarrow$ (ii). To verify that (ii) $\Rightarrow$ (iii), it suffices to do it for W of the form $i_P^G X$, with X in (Alg $L)(D)$. Since for any proper parabolic Q' of L, we have $r_{Q'}^L X = 0$, the decomposition 2.5 of $r_Q^G i_P^G$ into functors ir shows that $r_Q^G i_P^G X$ admits a composition series of successive quotients in (Alg $M)(E)$, for (M, E) conjugate to (L, D), or induced representations, to which we apply 2.4.

Let us prove that (iii) $\Rightarrow$ (iv). Applying 2.7 to L and to the component in (Alg L) (outside an $N_G(L)$-conjugate of D) of $r_P^G W$, we find that r_P^G is in the desired sum. Applying 2.7 to Ker ϕ, we find that ϕ is injective.

Lemma 0.8 (2.9)**.** *Soient X in (Alg $G)(L, D)$ and Y in (Alg $G)(M, E)$. If (L, D) and (M, E) are not conjugate, we have Hom$(X, Y) = 0$.*

Proof. Let $f : X \to Y$. Using 2.8 (iii), we deduce from 2.7 that $\mathrm{Im}(f) = 0$. □

Proposition 0.9 (2.10)**.** *The category (Alg G) is the direct sum, indexed by the (L, D) as in 2.6, taken up to conjugation, of the subcategories (Alg $G)(L, D)$.*

Given 2.9, it remains to show that every object of (Alg G) embeds into a sum $\bigoplus X_{L,D}$ with $X_{L,D}$ in (Alg $G)(L, D)$. Proceeding by induction, we can assume the theorem already known for the Levi subgroups of proper parabolics of G. Let W be a representation of G. We have

$$W = W_{qc} + W_{nqc} \tag{2.3.1}$$

with

$$W_{qc} \text{ in } (\text{Alg } G)_{qc} = \bigoplus (\text{Alg } G)(G, D), \text{ and} \tag{1.20.1}$$

$$W_{nqc} \text{ embedding into a sum of } i_P^G(X_P). \tag{2.5.1}$$

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The induction hypothesis allows for the decomposition of the X_ρ, and it only remains to use the transitivity of induction.

Proposition 0.10 (2.11)**.** *Let z be in the center of the abelian category $(Alg\ G)(L, D)$. For P a parabolic with Levi L and π in D, z acts as a scalar in $i_P^G(\pi)$. This scalar $z(\pi)$ is independent of P, and the function $\pi \mapsto z(\pi)$ is a regular function on the algebraic variety D.*

Let $M = L/L^0$, $B = \mathbb{C}[M]$ the affine algebra of the algebraic torus T with character group M, and $\chi_{\text{un}} : M \to B$ the "universal" character of M with values in B. Let (V, π) be an isomorphism class in D, and the induced representation

$$W = i_P^G(V \otimes B, \pi \otimes \chi_{\text{un}}).$$

This is a B-module, and for every $\mathbb{C}$-homomorphism $\sigma : B \to \mathbb{C}$ corresponding to $\chi : M \to \mathbb{C}^*$ (with $\chi = \sigma\chi_{\text{un}}$), we have

$$W \otimes_{B,\sigma} \mathbb{C} = i_P^G(V, \pi \otimes \chi).$$

For every open compact subgroup K, W^K is a projective B-module of finite type.

It is known that for a Zariski dense set of χ in $T(\mathbb{C}) = \text{Hom}(M, \mathbb{C}^*)$, $i_P^G(V, \pi \otimes \chi)$ is irreducible. This is for example the case for π_χ unitary and not isomorphic to its conjugate by a non-trivial element of $N(L)/L$. According to 1.17, there exists b in B such that z acts on W by multiplication by b. For each $\chi \in T(\mathbb{C})$, z thus acts in $i_P^G(\pi_\chi)$ by multiplication by $b(\chi)$. For η in the finite group F of characters of M such that $\pi\eta \sim \pi$, $i_P^G(\pi_\chi)$ is isomorphic to $i_P^G(\pi_{\chi\eta})$, whence $b(\chi\eta) = b(\chi)$. The function b thus descends to a regular function on $T/F \sim D$. It remains to verify the independence of P. It follows from the fact that for a Zariski dense set of χ (for example the same ones as above), the isomorphism class of $i_P^G(\pi_\chi)$ is independent of P.

2.12. Let $W(L, D)$ be the subgroup of $N(L)/L$ formed by the elements n such that D coincides with its conjugate by n. This finite group acts on D and the function $z(\pi)$ from 2.11 is invariant under $W(L, D)$: it is a regular function on the quotient algebraic variety.

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To each z in the center of $H(G)^\wedge$, i.e., in the center of the abelian category (Alg G), we thus associate a collection of regular functions on the $D/W(L, D)$ for (L, D) as in 2.6, i.e., a regular function on the variety disjoint union of the $D/W(L, D)$.

Theorem 0.11 (2.13)**.** *The construction above establishes an isomorphism of $Z(H(G)^\wedge)$ with the ring of regular functions on $\coprod D/W(L, D)$ (disjoint union over the (L, D) as in 2.6, taken up to conjugation).*

According to 2.10 and 1.9, it remains to show that for each (L, D), the map $z \mapsto z(\pi)$ identifies the center of the abelian category $(\text{Alg } G)_{(L,D)}$ with the ring of regular functions on $D/W(L, D)$. The injectivity follows from the fact that every irreducible of $(\text{Alg } G)_{(L,D)}$ is a subquotient of an $i_P^G(\pi)$, π in D. Let us prove the surjectivity. Given 1.15, applied to L, it suffices to prove the

Lemma 0.12 (2.14)**.** *For every element z_L of the center of $(Alg\ L)_{(D)}$, invariant by $W(L, D)$, there exists an element z of the center of $(Alg\ G)_{(L,D)}$ which on every induced representation $i_P^G X$ (X in $(Alg\ L)_{(D)}$, and P a parabolic with Levi L) acts as i_P^G(endomorphism of X defined by z_L).*

According to 2.8 (iv), we have, for every W in $(\text{Alg } G)_{(L,D)}$, an injection

$$\varphi : W \hookrightarrow \bigoplus_P i_P^G[(r_P^G W)_{(D)}]. \tag{2.14.1}$$

Lemma 0.13 (2.15)**.** *The image $\varphi(W)$ of φ is stable under $\bigoplus_P i_P^G(z_L)$.*

Proof that 2.15 $\Rightarrow$ 2.14. The endomorphism $\bigoplus_P i_P^G(z_L)$ induces on W a functorial endomorphism, which is the desired element of the center. □

Proof of 2.15. We leave it to the reader to verify that the set of W satisfying 2.15 is stable under subquotients and sums. This reduces us to the case where W is of the form $i_P^G X$, with X in $(\text{Alg } L)_{(D)}$. For such a W, we will show more precisely that for every Q with Levi L, the diagram □

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$$\begin{array}{ccc} i_P^G X & \xrightarrow{\alpha} & i_Q^G[r_Q^G i_P^G X(D)] \\ \Big\downarrow i_P^G(z_L) & & \Big\downarrow i_Q^G(z'_L) \\ i_P^G X & \xrightarrow{\alpha} & i_Q^G[r_Q^G i_P^G X(D)] \end{array}$$

is commutative. In this diagram, z_L denotes on the one hand the endomorphism of X, and on the other hand that of $(r_Q^G i_P^G X)(D)$, defined by z_L. By adjunction, it is equivalent to verify that:

Lemma 0.14 (2.16)**.** *The endomorphism z_L of $r_Q^G i_P^G(X)(D)$ coincides with the endomorphism z'_L deduced by functoriality from the endomorphism z_L of X.*

The decomposition 2.5 of ri into functors ir shows that $r_Q^G i_P^G(X)(D)$ admits a finite functorial filtration F, admitting the following $Gr_F^i = F^i/F^{i+1}$: they are indexed by $n \in W(L,D)$, and for $\tilde{n}$ a representative of n in $N(L)$, the $Gr_F^i r_Q^G i_P^G(X)(D)$ is isomorphic to the functor $(X,\pi) \longrightarrow \tilde{n}(X,\pi) = (X, \pi \circ \text{int}_{\tilde{n}}^{-1})$.

The symmetry of z_L by $W(L,D)$ ensures that on each of the $Gr_F^i r_Q^G i_P^G(X)(D)$, we have $z_L = z'_L$. This suffices to establish the lemma for those X such that, for any $n_1 \neq n_2$ in $W(L,D)$, with representatives $\tilde{n}_1$ and $\tilde{n}_2$, we have $\text{Hom}_{L,\text{End}(X)}(\tilde{n}_1(X), \tilde{n}_2(X)) = 0$.

The set of X in $(\text{Alg } L)(D)$ satisfying 2.16 is stable under sums and sub-quotients. In view of 1.18, it suffices therefore to treat the case where X is the representation $(V \otimes B, \pi \otimes \chi_{\text{un}})$ from the proof of 2.11. For this representation, we have $B \subset \text{End}(X)$. We are going to verify that $\text{Hom}_{L,B}(\tilde{n}_1(X), \tilde{n}_2(X)) = 0$. For any compact open subgroup K, $\tilde{n}_i(X)^K$ is a projective B-module of finite type. It is therefore sufficient to verify that for a Zariski-dense set of $\chi \in \text{Hom}(B, \mathbb{C}) = \text{Hom}(M, \mathbb{C}^*)$, we have

$$\text{Hom}_L(\tilde{n}_1(X) \otimes_{B,\chi} \mathbb{C}, \tilde{n}_2(X) \otimes_{B,\chi} \mathbb{C}) = 0$$

We have $\tilde{n}_i(X) \otimes_{B,\chi} \mathbb{C} = \tilde{n}_i(X \otimes_{B,\chi} \mathbb{C}) = \tilde{n}_i(V \otimes \chi)$, and it is therefore sufficient to take χ such that $\tilde{n}_1(V \otimes \chi)$ is not isomorphic to $\tilde{n}_2(V \otimes \chi)$. That most χ are suitable follows from the fact, already visible on the central characters,

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that $W(L,D)$ acts faithfully on D.

Remark 1 (2.17). Let f be a regular function on $\coprod D/W(L,D)$, and z_f the corresponding element of the center of $H(G)^\wedge$. For every isomorphism class of irreducible representation π, let us denote by $f(\pi)$ the scalar by which z_f acts in this representation. If π is one of the constituents of the induced representation $i_P^G(\sigma)$, with σ cuspidal, we have $f(\pi) = f(\sigma)$. Let us consider z_f as a distribution on G (1.6). For every open compact subgroup K, $z_f * e_K$ has compact support — in particular it is in the Schwartz space of G — and one can apply to it the Plancherel formula. One thus obtains the following integral formula for z_f. The integral is over the space $\hat{G}_t$ of tempered representations of G, dg is a Haar measure and $d\pi$ the corresponding Plancherel measure:

$$z_f = \int_{\hat{G}_t} f(\pi) \cdot (\Theta_\pi(g^{-1}) dg) d\pi$$

Example 0.1 (2.18)**.** Let H be a division algebra over F and $G = GL(n,H)$. The conjugacy classes of parabolics correspond to the ordered partitions of n, the Levi L of the parabolic corresponding to $(n_i)_{1 \leq i \leq m}$ being the product of the $GL(n_i, H)$ and $N(L)/L$ the subgroup of the symmetric group of $[1,m]$ respecting the function $i \mapsto n_i$ (a product of symmetric groups). Let, for each i, π_i be an isomorphism class of cuspidal representation of $GL(n_i, H)$, and let D_i be the orbit of π_i under torsion by an unramified character $\omega_t : h \mapsto t^{v(\text{Nrd } h)}$ $(t \in \mathbb{C}^*)$. If F_i is the subgroup of the group of $(n_i[H:F]^{1/2})$-th roots of unity such that $\pi_i \sim \pi_i \omega_t$ for $t \in F_i$, we have $\mathbb{C}^*/F_i \xrightarrow{\sim} D_i$. Let π be the representation $\bigotimes \pi_i$ of L, and $D = \prod D_i$ its orbit under torsion by a character of L/L°. The subgroup $W(L,D)$ of $N(L)/L$ is the subgroup of the symmetric group of $[1,m]$ respecting $i \mapsto (n_i, D_i)$; it is a product of symmetric groups, and its action on $D = \prod D_i$ is the obvious action. The quotient $D/W(L,D)$ is thus a non-singular algebraic variety.

Example 0.2 (2.19)**.** On the other hand, already for $G = SL(n, F)$, a quotient $D/W(L, D)$ can have singular points. Let us take $L =$ the group of diagonal matrices and $D =$ the set of characters of L/L°. It identifies with the quotient $(\mathbb{C}^*)^n/(\mathbb{C}^*$ diagonal), and $W(L, D)$ is the symmetric group, acting in the obvious way. Let ζ be a primitive n-th root of 1 and u the image in D of $(1, \zeta, \zeta^2, \dots) \in \mathbb{C}^{*n}$. The stabilizer of u

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in $W(L, D) \sim S(n)$ is the cyclic subgroup generated by the cycle $(1, \dots, n)$. For $n > 2$, the quotient $D/W(L, D)$ has an isolated singularity at the image of u.

3. APPLICATIONS: FINITENESS PROPERTIES.

3.1. For V a representation of G, we will have to consider the following finiteness properties:
V admissible: the V^K are finite-dimensional.
V of finite type: there exists a finite family of elements $v_i \in V$ $(i \in I)$ such that the gv_i $(g \in G, i \in I)$ linearly generate V.
For B a $\mathbb{C}$-algebra, and for V a B-module equipped with an action of G, we similarly define B-admissible (condition (B-adm) of 1.16) and B-of finite type.

In any case for B Noetherian, the functors i_P^G and r_P^G transform B-admissible representations (resp. B-of finite type) into B-admissible representations (resp. B-of finite type). For

$$\begin{array}{ll} i_P^G & \text{and } B\text{-admissible, and for} \\ r_P^G & \text{and } B\text{-of finite type,} \end{array}$$

it is an easy consequence of the compactness of G/P (see [1] 3.13). The two other cases will be treated in 3.11 and 3.5.3.

3.2. A representation V of finite type (even B-of finite type) has non-zero components only in a finite number of $(Alg\ G)(L, D)$: in the decomposition $V = \bigoplus V(D, L)$, each generator has only a finite number of non-zero coordinates. We can specify the (D, L) that appear in terms of an open compact subgroup fixing the generators.

Indeed, let K be an open compact subgroup of G, and (L, D). Let us show that the following conditions (3.2.1), (3.2.2) are equivalent.

(3.2.1). For all W in $(Alg\ G)(L, D)$, we have $W^K = 0$.

Let P be a parabolic with Levi L and W an irreducible representation of L, of isomorphism class in D. For any conjugate K' of K, let K'_L be the projection into L of $K' \cap P$. Up to conjugation, $(gKg^{-1})_L$ depends only on the double class of g in $P\backslash G/K$.

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(3.2.2). For each conjugate K' of K, $W^{K'_L} = 0$.

$$\text{For } W \text{ a representation of } L, \text{ one has } (i_P^G W)^K \sim \bigoplus_{P\backslash G/K} W^{K'_L}.$$

The condition (3.2.2) is thus equivalent to $(i_P^G W)^K = 0$. In particular, (3.2.1) $\Rightarrow$ (3.2.2). Since $K'_L \subset L^o$, it is independent of W, and ensures that for any representation W in $(\mathrm{Alg}\, L)(D)$, one still has $(i_P^G W)^K = 0$. For W irreducible, the isomorphism class of $i_P^G W$ does not depend in general on the Levi P of L. The condition (3.2.2) is thus independent of P, and one has (3.2.2) $\Rightarrow$ (3.2.1).

We will denote by $3(G)$ the center $H(G)^\wedge$, the product of the centers $3(G)(L, D)$ of the categories $(\mathrm{Alg}\, G)(L, D)$. If a representation V of G has no non-zero component except in the $(\mathrm{Alg}\, G)(L_i, D_i)$ $(i \in I$, assumed finite), $3(G)$ acts on V via its quotient $\prod_i 3(G)(L_i, D_i)$. Such a quotient is a $\mathbb{C}$-algebra of finite type, and in particular Noetherian.

Proposition 0.15. *3.3. Every representation of finite type is $3(G)$-admissible.*

Proof. We reduce to assuming the representation V is in an $(\operatorname{Alg} G)(L, D)$, and to proving that V is $3(G)(L, D)$-admissible. Let $3 = 3(G)(L, D)$.

Let us first treat the case where $L = G$: there then exists a cuspidal representation (V^o, π^o) of G^o such that $V|G^o$ is a sum of copies of G-conjugates of (V^o, π^o). Proceeding as in 1.12, with $M = G/G^o$, one writes

$$V = \operatorname{Ind}_{G_1}^G (V^o \otimes X),$$

X being a $\mathbb{C}[\tilde{M}_1; z^{-1}]$-module (1.13). For V to be of finite type, it is necessary and sufficient that the module X be so. Since $\mathbb{C}[\tilde{M}_1; z^{-1}]$ is of finite rank over its center Z, X is of finite type over Z. This center is none other than 3. Because V^o is admissible, and X of finite type over Z, the representation $V^o \otimes X$ of G_1 is Z-admissible. Finally, admissibility is preserved by $\operatorname{Ind}_{G_1}^G$.

General case. By definition of $(\operatorname{Alg} G)(L, D)$, V is contained in a sum of representations

$$V \hookrightarrow \bigoplus_P i_P^G X_P,$$

where P runs through the parabolics of Levi L and where X_P is in $(\operatorname{Alg})(D)$. □

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Since V is of finite type, one can take the X_p to be of finite type. The X_p are $3(L)(L, D)$-admissible, and thus 3-admissible, since $3(L)(L, D)$ is a module of finite type over 3. One indeed obtains 3 by passing to invariants under a finite group. Because i_p^G preserves admissibility, $\bigoplus_p i_p^G X_p$ is $3(G)(L, D)$-admissible, and V is so as well, the ring 3 being Noetherian.

Variant 3.3.1. Let B be a Noetherian $\mathbb{C}$-algebra. If G acts on a B-module V, and V is of finite type over B, then V is $3(G) \otimes B$-admissible. The proof is the same, given that $3(G)(L, D) \otimes B$ is a quotient of a polynomial ring over B, and therefore is a Noetherian ring.

Corollary 0.16 (3.4)**.** *For every open compact subgroup K of G, the Hecke algebra $H(G, K)$ is a module of finite type over $3(G)$.*

Proof. Apply 3.3 to the regular representation of G on $L_c^\infty(G/K)$ (compactly supported functions on G, right-invariant by K). It is generated by one generator: the characteristic function of K. We have $L_c^\infty(G/K)^K \simeq H(G, K)$.

In particular, $H(G, K)$ is a module of finite type over its center (which contains $e_K 3(G) e_K$), and the latter is a $\mathbb{C}$-algebra of finite type. One can prove more precise results — see 3.13, 3.14.

If we work with $\mathbb{Q}$ — rather than $\mathbb{C}$ — as the field of scalars, it follows from the complex case that $H_{\mathbb{Q}}(G, K)$ is still of finite type over its center, and that the latter is a $\mathbb{Q}$-algebra of finite type. It can be deduced that every irreducible $H_{\mathbb{Q}}(G, K)$-module is of finite dimension over $\mathbb{Q}$, and that every irreducible $\mathbb{Q}$-representation of G is admissible, with its commutant being a field of finite degree over $\mathbb{Q}$. Everything that has been said with $\mathbb{C}$ as the field of scalars can be restated with $\mathbb{C}$ replaced by any algebraically closed field of characteristic 0. □

3.5. Let P and $\overline{P}$ be two opposite parabolics, with common Levi L and respective unipotent radicals U and $\overline{U}$. Let K be an open compact subgroup, and consider the condition of 2.1 b):

$$[K] = [K \cap U] * [K \cap L] * [K \cap \overline{U}]. \tag{3.5.1}$$

Proposition 0.17 (3.5.2)**.** *If condition (3.5.1) is satisfied, for every representation V of G, we have*

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$$V^K \longrightarrow (V_U)^{K\cap L}$$

By passing to the inductive limit, we are reduced to assuming V is of finite type. The representation of L on the coinvariants V_U is still of finite type (3.1) and $(V_U)^{K\cap L}$ is a $\mathfrak{Z}(L)$-module[5] of finite type. We deduce that it is a $\mathfrak{Z}(G)$-module of finite type (for the module structure induced from that of V). Let $(v_i)_{i\in I}$ be a finite set of generators and lift each $v_i \in V_U$ to $\tilde{v}_i \in V^L$. For any a in the center of L, a acts by automorphism on V_U, and commutes with $\mathfrak{Z}(L)$. Thus the av_i generate V_U. They are the images of $a\tilde{v}_i$. Each v_i is fixed by an open subgroup of $\overline{U}$; if a is sufficiently expanding on $\overline{U}$, each av_i will be fixed by $[K\cap\overline{U}]$, and the condition (3.5.1) ensures that $w_i := [K\cap U]a\tilde{v}_i$ is fixed by K. The w_i still have av_i as image in V_U, and the $\mathfrak{Z}(G)$-module generated by the w_i — contained in V^K — maps onto $(V_U)^{K\cap L}$.

Remark 2 (3.5.3). If V is B-admissible, we deduce from 3.5.2 that $r_p^G(V)$ is also.

Reformulation 3.6. Let us take for V the left regular representation of G on the space $L_c(G)$ of locally constant functions with compact support. The map $L_c(G) \to L_c(U\backslash G)$:

$$f \mapsto \int_U f(ug)\,du$$

identifies $L_c(U\backslash G)$ with the quotient $L_c(G)_U$ of $L_c(G)$. We find that for every function $f \in L_c(U\cdot(L\cap K)\backslash G)$, there exists a function F in $L_c(K\backslash G)$ such that

$$f(g) = \int_U F(ug)\,du.$$

It would not be difficult to deduce 3.5.2 from this particular case.

3.7. It follows from 3.5.2 that there exist arbitrarily small open compact subgroups K satisfying the two following conditions:

(3.7.1) Let L be a Levi subgroup. For $P = LU$ a parabolic with Levi L and K' a conjugate of K, the conjugacy class in L of $K'_L = (K'\cap P)/(K'\cap U)$ is independent of P and K'.

(3.7.2) For every parabolic $P = LU$ and every representation V of

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G, we have

$$V^K \longrightarrow (V_U)^{K_L}$$

Indeed, let A be a maximal split torus and K_o such that for every parabolic P one has $G = K_oP$. If K is invariant in K_o, the conditions (3.7.1) and (3.7.2) are implied by (3.5.1) for every pair of opposite parabolics $P, \overline{P}$ containing A. Indeed:

a) It suffices to verify (3.7.1) for $L \supset A$. Let us set $K' = (pk_o)K(pk_o)^{-1} = pKp^{-1}$; we find that K'_L is conjugate to $K\cap L$. For (3.7.2), we similarly reduce to assuming that $P \supset A$, and we apply 3.8. Arguments 2.1 b) provide the existence of K as soon as there exists in the Lie algebra $\mathfrak{g}$ of G a lattice Λ stable under K_o and the sum of its intersections with the subspaces $\mathfrak{g}_\alpha$, of weights of A in $\mathfrak{g}$. Such a lattice exists[6] if K_o is defined by a special point of the apartment defined by A.

For $G = \mathrm{GL}(n, F)$, one can take for K the congruence subgroup of level $m \geq 1$: $K = \{k \mid k \equiv 1(\mathfrak{p}^m)\}$.

Proposition 0.18 (3.8)**.** *Let V be a representation of G, and K as in 3.7. For V to be generated by V^K, it is necessary and sufficient that, for all (L, D), the following condition is satisfied. Let W be a representation of L of isomorphism class in D; if $W^{K_L} = 0$, the component of V in $(\mathrm{Alg}\,G)(L, D)$ is zero.*

We reduce to assuming V is in an $(\mathrm{Alg}\,G)(L, D)$. Let W be a representation of L, of isomorphism class in D. Let us distinguish two cases.

Case 1. $W^{K_L} = 0$. Under this hypothesis, according to 3.2, we have $V^K = 0$ for all V in $(\mathrm{Alg}\,G)(L, D)$.

[5] The symbol appearing as a script '3' in the original text is the standard notation for the center of the universal enveloping algebra or Hecke algebra, typically denoted $\mathfrak{Z}$.

[6] [[UNCERTAIN: existe — CONFIDENCE: 95% — REASON: The word is slightly faint but clearly 'existe' in context.]]

Case 2. $W^{K_L} \neq 0$. Let V' be the quotient of V by the subrepresentation generated by V^K. We must show that $V' = 0$. In other words, it suffices to show that if $V^K = 0$, then $V = 0$. The hypothesis $V^K = 0$ ensures that for every parabolic P containing L, $[(r_P^G V)(D)]^{K_L} = 0$. Since $W^{K_L} \neq 0$, it follows that $r_P^G(V)(D) = 0$. That $V = 0$ follows by .

Corollary 0.19 (3.9)**.** *(i) For K as in 3.7, the category of representations V of G generated by V^K is the sum of a finite number of* $(\mathrm{Alg}\, G)(L, D)$.

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In particular, it is stable under sub-quotients.
(ii) The functor $V \mapsto V^K$ is an equivalence from this category to that of $H(G, K)$-modules.
Assertion (i) results from 3.8 and from the finiteness theorem [1] §4 already used (cf. 2.3).
For any K, the functor $V \mapsto V^K$ has as a left adjoint

$$I : W \mapsto (H(G)e_K) \otimes_{H(G,K)} W.$$

We have $W \xrightarrow{\sim} I(W)^K$; the map $I(V^K) \to V$ is surjective if V is generated by V^K, and its kernel has no non-trivial vector fixed under K. By (i), it is zero, and (ii) follows.

Corollary 0.20 (3.10)**.** *For a representation V of G to be of finite type, it is necessary and sufficient that it be $\mathfrak{Z}(G)$-admissible, and that it has non-zero components only in a finite number of* $(Alg\ G)(L, D)$.

Necessity is 3.3. Sufficiency: by 3.8, V is generated by V^K for a suitable K, which is itself of finite type under $\mathfrak{Z}(G)$ — *a fortiori* under $H(G, K)$.

Variant 3.11. If B is a Noetherian $\mathbb{C}$-algebra, similarly for B-of finite type, with $\mathfrak{Z}(G)$ replaced by $B \otimes \mathfrak{Z}(G)$. Since i_P^G preserves admissibility, it follows from 3.10 that the functor i_P^G preserves the "finite type" condition. Similarly for B-of finite type (for Noetherian B).

Remark 3.12. It follows from 3.10 that a subrepresentation of a representation of finite type is again of finite type: the category of these representations is Noetherian ([1] 4.19). The representations of finite length are those which are both of finite type and admissible.

3.13. Let K be an open compact subgroup. The Hecke algebra $H(G, K)$ is the product of the $H(G, K)(L, D) := e(L, D)H(G, K)e(L, D)$, where $e(L, D)$ is the central idempotent that projects each representation onto its component in $(\mathrm{Alg}\ G)(L, D)$. It suffices to extend the product over the (L, D), in finite number, not satisfying (3.2.1) (3.2.2).

Each $H(G, K)(L, D)$ is a module of finite type over $\mathfrak{Z}(G)(L, D)$. One can specify its structure:

Proposition 0.21 (3.14)**.** *Let W be a representation of L of class in D,*

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P a parabolic with Levi L and $N = \dim(i_P^G(W)^K)$: Assume that $N \neq 0$.

(i) $\mathfrak{Z}(G)(L, D)$ is the center of $H(G, K)(L, D)$, and $H(G, K)(L, D)$ is a torsion-free $\mathfrak{Z}(G)(L, D)$-module.

(ii) There exists $f \neq 0$ in $\mathfrak{Z}(G)(L, D)$ such that the algebra $H(G, K)(L, D)[1/f]$ obtained from $H(G, K)(L, D)$ by extension of scalars from $\mathfrak{Z}(G)(L, D)$ to $\mathfrak{Z}(G)(L, D)[1/f]$ is an Azumaya algebra of rank N^2 over $\mathfrak{Z}(G)(L, D)$.

Let $H := H(G,K)(L,D)$ and $\mathfrak{Z} := \mathfrak{Z}(G)(L,D)$. Let $B = \mathbb{C}[L/L^\circ]$ be the affine algebra of the torus with character group L/L° and, as in [ ][7], the representation $W \otimes \chi_{\mathrm{un}}$ of L in $W_B := W \otimes B$. By inducing, one obtains a representation $i_P^G(W \otimes \chi_{\mathrm{un}})$ of G, equipped with a B-module structure. Passing to the invariants under K, one obtains an $H, B - B$-module U. As a B-module, it is free of rank N. We will use the following facts:

(a) The H-module U is faithful.

If h acts trivially, it acts trivially in each $U \otimes_B \mathbb{C}$ for every homomorphism $\sigma : B \to \mathbb{C}$, therefore in each V^K for V irreducible in (Alg $G)(L,D)$. We then have $h = 0$.

(b) $\mathfrak{Z}$ is obtained from B by passing to the invariants under a finite group (and the two actions on U coincide).

(c) H is of finite type over $\mathfrak{Z}$.

(d) For a Zariski-dense set of $\omega \in T(\mathbb{C}) = \mathrm{Hom}(B,\mathbb{C})$, $U \otimes_{B,\omega} \mathbb{C}$ is a simple H-module: $H \to \mathrm{End}_{\mathbb{C}}(U \otimes_{B,\omega} \mathbb{C})$ and its isomorphism class depends only on its restriction to $\mathfrak{Z}$.

Indeed, $i_P^G(W \otimes \chi_{\mathrm{un}}) \otimes_{B,\omega} \mathbb{C}$ is irreducible, with the desired independence of ω.
For a Zariski-dense set of $\alpha : \mathfrak{Z} \to \mathbb{C}$, the diagram

$$\begin{array}{ccc} H & \xrightarrow{(a)} & \mathrm{End}_B(U) \\ \vdots & & \vdots \\ \mathfrak{Z} & \longrightarrow & B \end{array}$$

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gives rise (thanks to (b)(c)) to

$$H \otimes_{\mathfrak{Z},\sigma} \mathbb{C} \hookrightarrow \prod_{\tilde{\sigma}} \mathrm{End}_{\mathbb{C}}(U \otimes_{B,\tilde{\sigma}} \mathbb{C})$$

(product over the $\tilde{\sigma}$ extending σ), which by (i) factors through a bijection onto a diagonal $M_N(\mathbb{C})$. Assertion (ii) follows from this. The "torsion-free" assertion results from (a), and the fact that $\mathfrak{Z}$ is the center of H follows from $\mathfrak{Z}$ being integrally closed.

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